

Bangaltali and Rupakari Watershed

R-65

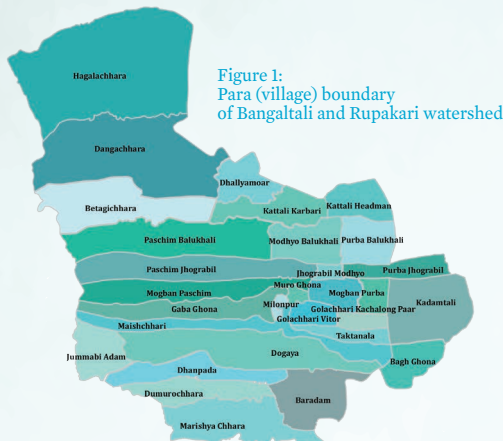


Figure 1:
Para (village) boundary
of Bangaltali and Rupakari watershed

Source:
Center for Environmental and
Geographic Information Services (CEGIS)

Disclaimer:
Boundaries are not
necessarily authoritative

The Bangaltali and Rupakari sub-watershed (R-65), located in the Bangaltali, Marishya, and Rupakari Unions, 376 no Tintila, 377 no Rupakari, and 378 no Marishya of the Baghaichhara Upazila of the Rangamati District, encompasses 30 paras (figure 1) and covers approximately 3,401 hectares of land. It is home to over 8,000 people, with communities consisting of the Chakma ethnic group only. The overall literacy rate is approximately 92%, yet opportunities for higher education are very limited. Farming is the main occupation. The average monthly income is 16,914 taka (\$141), which is relatively low compared to the national average. About 81% of the houses are katcha and almost 38% of households use solar electricity systems. There is a noticeable scarcity of drinking water and a prevalent incidence of waterborne diseases.

Topography

Only 4% area towards the western region lies between 101-120-meter (m) elevation (figure 2). The west face of the watershed has comparatively higher elevation. Only 11% of the area lies above 120m elevation, which is situated in the western region of the watershed. Most of the study area lies between 40-60m (31%) and 61-80m (22%) elevation in the central region of the watershed. About 20% of the area lies below 40m, which consists mostly of flat croplands, and the rest of the 12% of the area lies between 81-100m elevation in the watershed.

Figure 2: Topography

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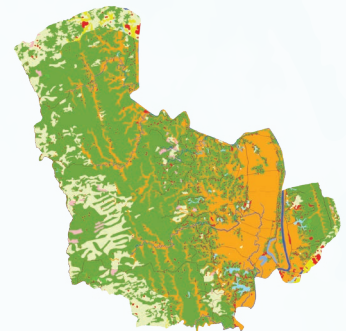
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Landuse

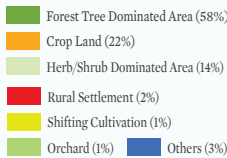
A total 13 types of land use have been identified within the R-65 watershed (figure 3). The main land use is forest (1,963 hectares (ha)), followed by cropland (760 ha), herb/shrub (453 ha), and agricultural areas (813 ha).

Figure 3: Land use



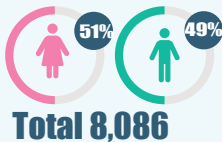
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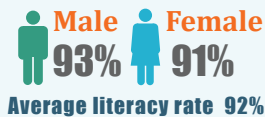


Demography

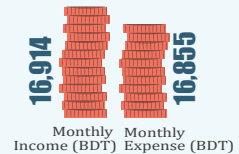
Population



Literacy



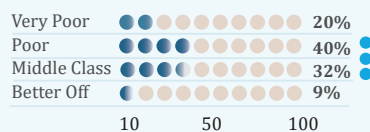
Economy



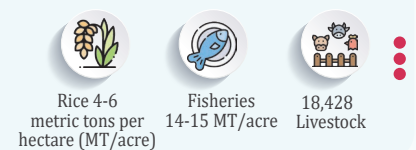
WASH



Poverty



Production



Climatic Hazards



Climate Change-Induced Hazards

Eight climate hazards threaten Bangaltali and Rupakari watershed agriculture, livelihood, ecosystem, and biodiversity. Drought, erratic rainfall, flash floods, heat waves, river floods, and lightning intensify across most paras. Landslides and hailstorms occur at low to moderate intensity.

Climate Change Risk

Climate change risk assessment identified 7 paras as very high-risk hotspots, 18 as high-risk, as medium-risk, 1 as low-risk, and the remaining para as very low-risk (Figure 4).

Figure 4: Mutihazard risk map



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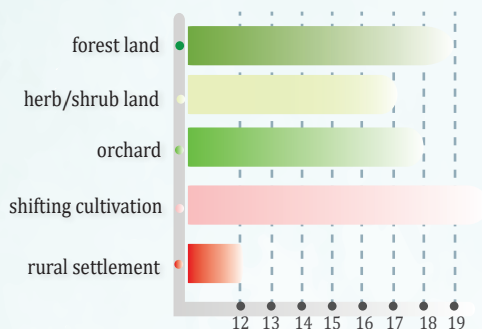
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Watershed Degradation

Land Degradation Status

In R-65, land degradation is prominent in -



Causes of Degradation

- Jhum cultivation with short rotation
- Improper crop management
- Overgrazing
- Deforestation & overharvesting of trees
- Lack of vegetation cover
- Climate change
- Overuse of inorganic fertilizers

Degradation Triggering Factors

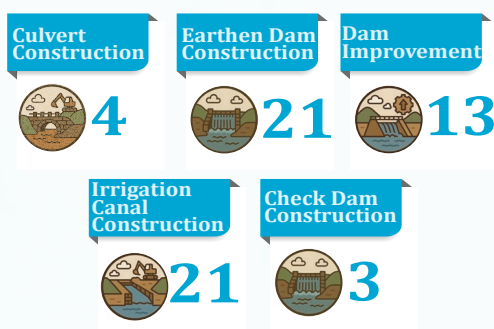
- Reduction of vegetation cover
- Quality and species composition change/diversity decline/monoculture plantation
- Soil erosion/landslides
- Surface sealing/hard pan development
- Waterlogging
- River bank erosion

Watershed Conservation

Conservation measures focused on enhancing ecological resilience as well as improving livelihood quality of the local community were identified and categorized into 6 classes.



Watershed Management Infrastructures



Watershed conservation interventions in Bangaltali and Rupakari include dam construction (21), irrigation canal construction (21), and dam improvement (13) being the most implemented, aimed at enhancing water storage and distribution. In addition to these, other conservation measures have been identified to further boost water availability, agricultural productivity, and overall ecosystem health. Measures such as check dam construction, culvert construction, and protection wall construction will help control runoff, reduce soil erosion, and protect critical infrastructure. Together, these interventions will significantly benefit the local community by ensuring year-round water access, improving crop yields, and reducing vulnerability to climate-induced hazards. These integrated efforts promote sustainable livelihoods and strengthen resilience in the Bangaltali and Rupakari region.